

Current Application Flow And Planned Oracle Enterprise AI Stack

A sendable explanation of what the application is doing today, what Oracle enterprise AI capabilities exist, and what the future state would look like.

Purpose Of This Document

- This document explains the current application in plain language for a reader who does not know the code.
It covers three things:
- ? The complete current flow and the technology stack used for each part
 - ? The Oracle enterprise AI capabilities documented in the reference material
 - ? Which Oracle-managed capabilities can replace current custom components and what the planned future flow would look like

High-Level Summary

What The Application Does Today

- ? General question answering
- ? OCR on uploaded documents and images
- ? Document-based question answering
- ? Document-to-document comparison
- ? Web fallback when internal retrieval is weak

What It Is Built On Today

- ? Custom workflow orchestration
- ? Specialist AI services
- ? Tool servers for OCR, retrieval, and web search
- ? OCI-hosted models for LLM and embeddings
- ? Oracle database vector search
- ? Discovery and identity verification for trusted specialist services

Reader Takeaway

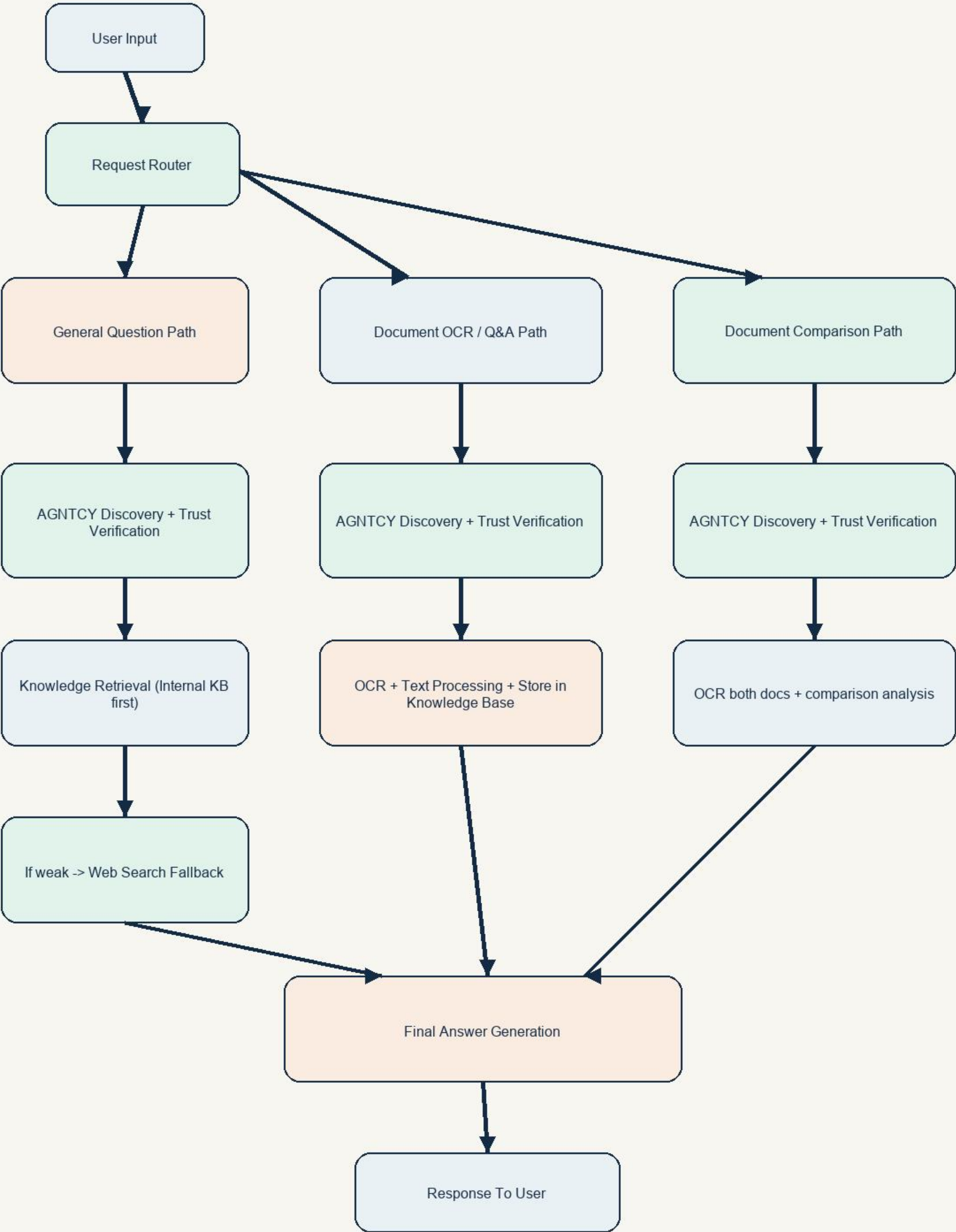
One-Sentence Framing

The current solution is a custom-built enterprise AI workflow running on OCI models and Oracle vector storage.
The planned direction is to keep the business experience but replace large parts of the custom AI runtime with Oracle's managed enterprise agent capabilities.

Current Application Flow

How requests move through the application today.

Current End-To-End Flow



How To Read This Flow

The router decides whether the request is a normal question, a document task, or a comparison task. Before a specialist agent is used, the application performs AGNTCY-based discovery and trust verification.

Current Technology Stack

What the application is using today, and what each layer is used for.

Stack Breakdown

Workflow Layer

- ? A workflow orchestrator decides what kind of request came in.
- ? It controls the sequence of steps for question answering, document processing, and comparison.
- ? It decides when to use internal retrieval and when to use web fallback.

Specialist AI Service Layer

- ? Focused specialist services handle document ingestion, knowledge retrieval, web search, and document comparison.
- ? The router delegates to these services depending on the request type.

Tooling And Data Layer

- ? OCR services extract text from PDFs and images.
- ? Text is chunked and converted into embeddings.
- ? An Oracle database stores the chunked text and vectors.
- ? Vector similarity search retrieves the most relevant content for answering.
- ? External search is used when internal retrieval is not strong enough.

Model Layer

- ? OCI-hosted foundation models are used for final answer generation.
- ? OCI-hosted embedding models are used for the knowledge retrieval pipeline.
- ? OCI-hosted models are also used for document comparison summaries.

Trust And Discovery Layer

- ? A discovery layer is used to find the right specialist service dynamically based on capability.
- ? An identity-verification layer checks that the discovered specialist service is trusted before it is used.
- ? This makes the architecture more enterprise-ready than hard-coding every endpoint directly.

Simple Summary

Today's application is a custom enterprise AI stack built from custom orchestration + specialist services + tool servers + OCI models + Oracle vector search.

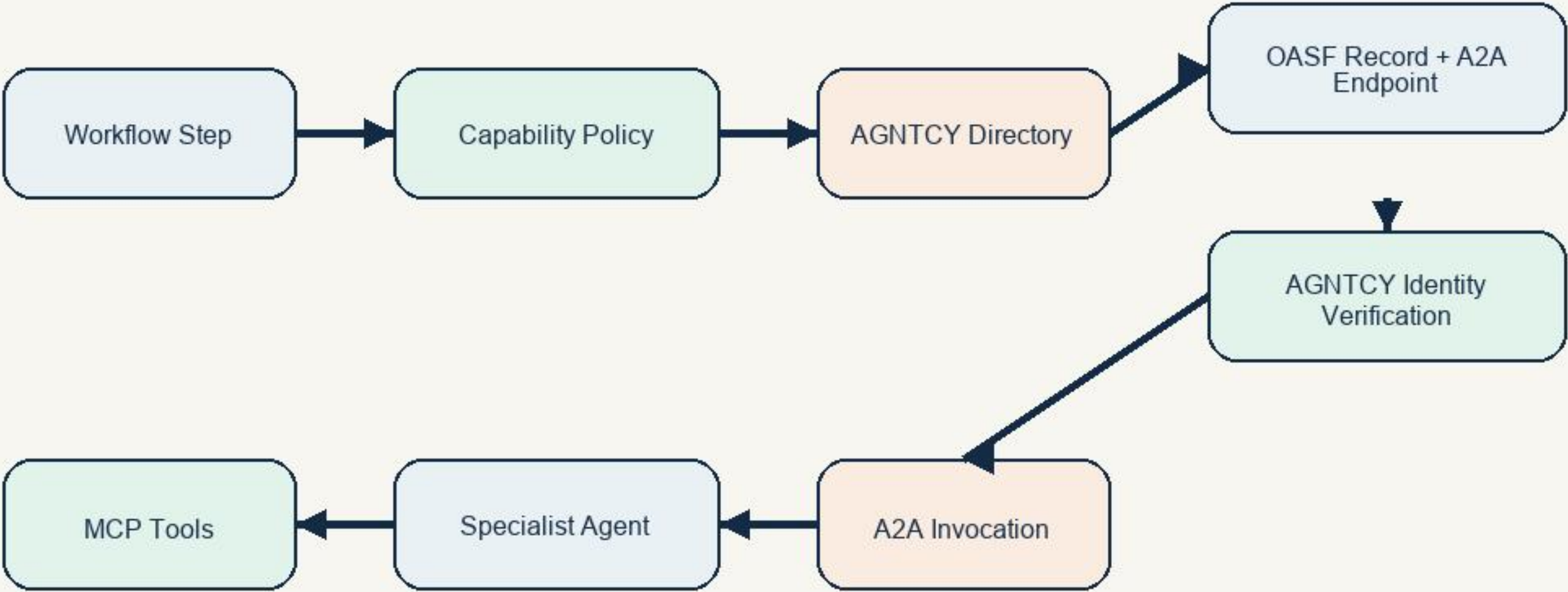
How AGNTCY Is Used In The Application

AGNTCY is a real runtime layer in this application, not only a setup-time or metadata-only addition.

AGNTCY Runtime Role

One-Line Description

This application uses AGNTCY as the metadata, discovery, and identity-trust layer on top of an A2A + MCP multi-agent runtime. AGNTCY does not replace the actual agent-to-agent messaging protocol or the tool execution layer. Instead, it decides which specialist should be trusted and used for a given capability.



What AGNTCY Is Doing Here

- ? The application maps each workflow phase to a capability policy, such as document indexing, KB question answering, web search, or comparison-oriented synthesis.
- ? AGNTCY Directory is queried to find a matching specialist A2A agent record based on those skills.
- ? The application reads the A2A endpoint and the identity metadata from the discovered OASF record.
- ? AGNTCY Identity is then used as a runtime gate: if badge verification fails, the application does not trust or invoke that agent.
- ? Only after discovery and trust verification does the application invoke the specialist agent over A2A.
- ? The specialist agent still performs its real work through its own logic and tool access, often using MCP servers underneath.

AGNTCY Components Used In The Application

How Directory, OASF, Identity, and A2A fit together in the current design.

AGNTCY Stack In This App

OASF Records

- ? Describe each specialist runtime component
- ? Carry skills metadata
- ? Carry A2A integration data
- ? Carry identity annotations

AGNTCY Directory

- ? Searches over capability metadata
- ? Returns matching agent records
- ? Acts as the main runtime discovery source

AGNTCY Identity

- ? Fetches and verifies agent badges
- ? Validates trust before the agent is used
- ? Works as a runtime trust gate

A2A

- ? Actual agent-to-agent communication protocol
- ? Used only after AGNTCY discovery and verification succeed

Official AGNTCY Framing

How This Matches The Official AGNTCY Model

- ? AGNTCY Directory is documented as a capability-based directory that stores metadata records and enables discovery of agent applications by their skills.
- ? OASF is documented as the standardized schema framework used to describe agents, MCP servers, and MAS metadata for interoperability and discovery.
- ? AGNTCY Identity is documented as the service used to verify identities and manage digital identities for agentic services, including A2A agents, MCP servers, and OASF-defined services.
- ? In this application, those official concepts are used in a real runtime chain: OASF records describe the agents, Directory discovers them by skill, Identity verifies them, and A2A is used to communicate with them.

What AGNTCY Is Not Doing Here

- ? It is not replacing A2A messaging itself.
 - ? It is not directly performing OCR, retrieval, or web search.
 - ? It is not replacing the business workflow layer.
- Its role is metadata, capability discovery, and trust verification.

Oracle Enterprise AI Capabilities Mentioned In The Reference

The managed capabilities available in Oracle's documented agent platform.

Documented Capability Groups

Governance And Runtime

- ? Projects
- ? IAM and authentication
- ? OCI Responses API
- ? Conversations / managed multi-turn state
- ? Streaming
- ? Structured outputs
- ? Reasoning controls
- ? Multimodal input

Memory

- ? Short-term memory
- ? Long-term memory
- ? Memory compaction
- ? Managed continuity within a project and across future sessions

Tools And Data

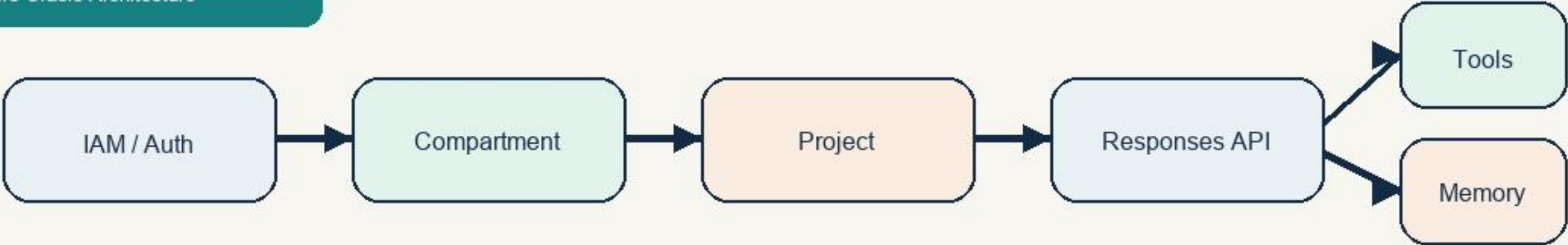
- ? File Search
- ? Code Interpreter
- ? Function Calling
- ? Remote MCP Calling
- ? SQL Search / NL2SQL
- ? Vector Stores
- ? Files API
- ? Connectors
- ? Containers and Container Files

Important Interpretation

What This Means

Oracle provides a managed enterprise agent platform.
The current application is already enterprise-oriented, but it implements much of the runtime behavior manually.
The opportunity is to keep the business workflows while replacing custom AI runtime pieces with managed Oracle capabilities.

Core Oracle Architecture



What Oracle Capabilities Can Replace In The Current Application

Direct mapping from the current custom stack to Oracle-managed enterprise AI features.

Replacement Map

Current Custom Capability	Oracle Enterprise AI Replacement
Custom final response composition	OCI Responses API
Custom multi-turn handling	Conversations API + short-term memory
Custom retrieval over OCR content	Files API + Vector Stores + File Search
Custom app-side tool orchestration	Function Calling
Custom external tool integration	Remote MCP Calling
No managed cross-session memory	Long-term memory
Custom execution-style processing	Code Interpreter + Containers
Future natural-language-to-database capability	SQL Search / NL2SQL

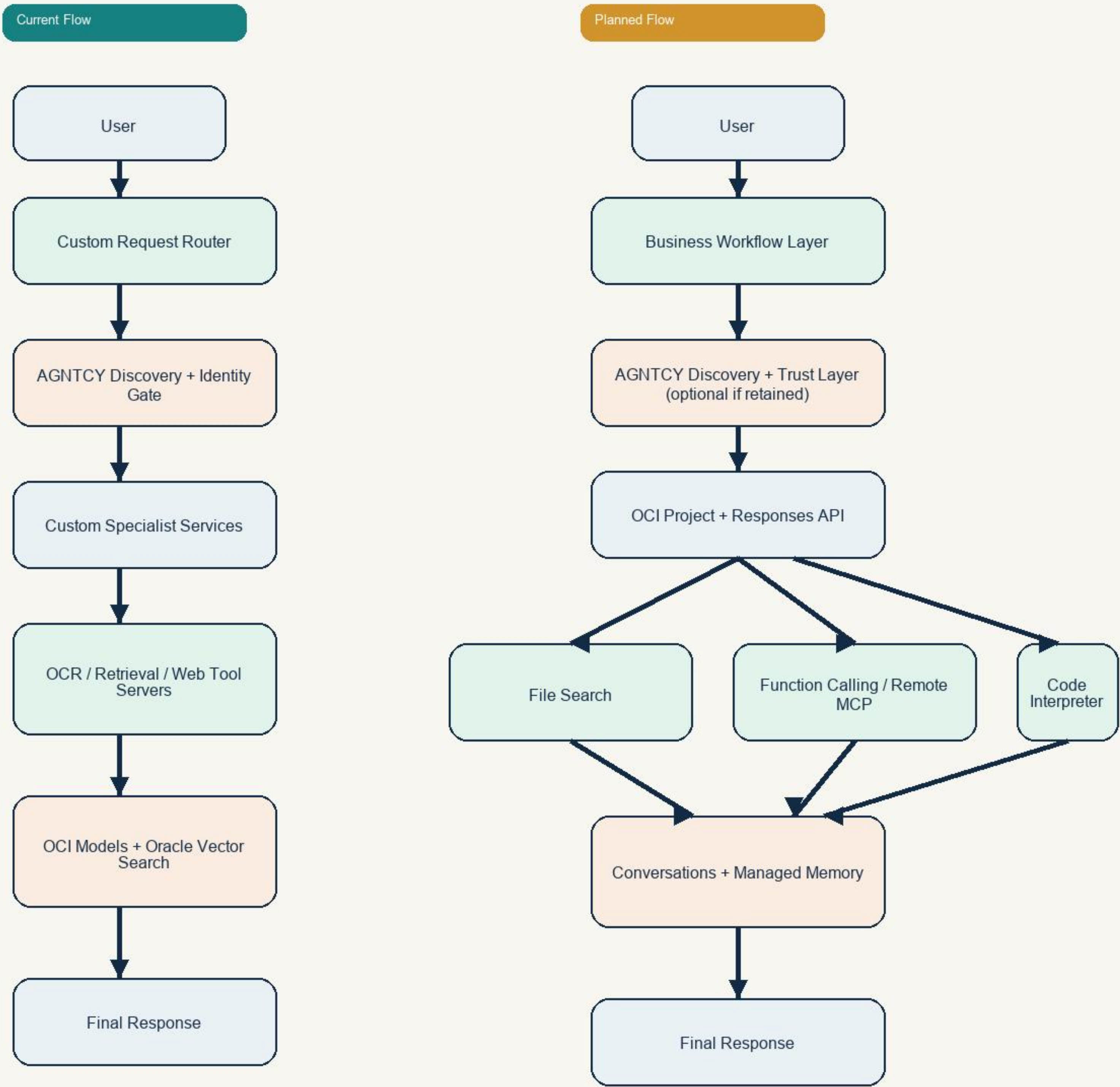
What Should Still Remain

Recommended Principle

- ? Keep the business workflow layer that understands the product experience and request types.
- ? Keep the document-centric experience: OCR flow, document Q&A flow, and comparison flow.
- ? Keep discovery and trust layers if dynamic specialist-service discovery remains important to the wider architecture.
- ? Replace the low-level AI runtime plumbing underneath those workflows with Oracle-managed platform capabilities.

Current Flow Vs Planned Flow

A side-by-side view of what exists now and what the future Oracle-centered stack would look like.



How To Explain The Difference

Current state: the application already provides enterprise-style AI functionality, but much of the runtime orchestration, retrieval, and tool integration is custom-built, with AGNTCY used as the discovery and trust layer before specialist invocation.

Planned state: the business workflow remains, and AGNTCY can remain as the discovery/trust layer if desired, while the underlying AI runtime moves toward Oracle-managed Projects, Responses API, managed retrieval, managed tool calling, and managed memory.

This reduces custom AI plumbing while keeping the product experience and business-specific orchestration logic intact.